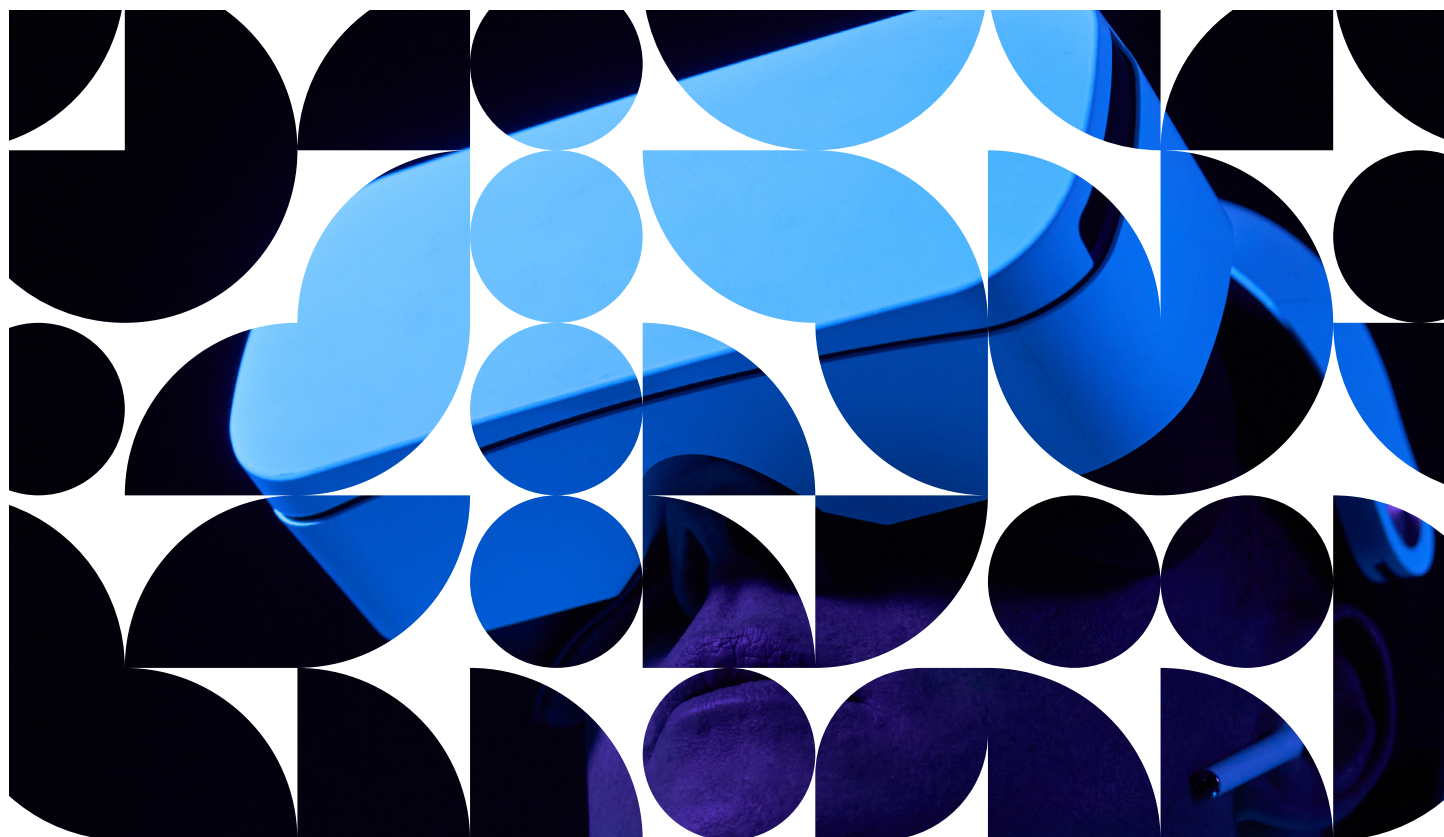


WHITE PAPER

Collaborative Work in Virtual Reality: Insights and Recommendations on Using Immersive Technologies in Workplace Contexts

Marco Dehnert, University of Arkansas
Matthew J. A. Craig, Central Michigan University
Maggie Liao, University of Georgia
Mia Wang, University of Colorado Boulder
Josh Baldwin, Boston University
Grace Ahn, University of Georgia
Bree McEwan, University of Toronto Mississauga



EXECUTIVE SUMMARY

Virtual reality (VR) technologies are rapidly moving beyond entertainment and experimental use cases toward becoming viable infrastructures for everyday collaboration. Organizations across sectors are exploring VR as a tool for remote teamwork, training, design, onboarding, and social connection among distributed employees. Advances in headset accessibility, real-time rendering, spatial audio, and avatar-based interaction have made immersive collaboration technically feasible at a moment when many workplaces are reevaluating how, where, and with whom work occurs (Doering & Tilcsik, 2025; Paradis & Altchek, 2025; Parker, 2025; Shumway, 2026; Stechyson, 2025). As hybrid and remote work models continue to evolve (and change in adoption), against the backdrop are the shadows of what was framed as a potential “next generation” collaborative environment—one that promises to move beyond traditional video conferencing by supporting embodiment, co-presence, and shared spatial experiences.

Despite the original enthusiasm for the metaverse and collaborative work in VR, significant uncertainty remains surrounding how collaborative work in VR should be implemented and evaluated. Existing workplace communication tools, such as email, messaging platforms (e.g., Teams, Slack), and video conferencing tools, have evolved over decades through social adaptation and policy development. By contrast, collaborative VR environments introduce fundamentally different interaction dynamics, including embodied presence, how people represent themselves through embodied avatars, continuous behavioral data capture (e.g., workplace surveillance), and new forms of spatial and social organization (e.g., re-representation of nonverbal cues such as proxemics, chronemics, haptics). Moreover, virtual environments for collaborative work are not a monolith, with differences in the affordances of technology. These changes raise important questions that are not purely technical but social, ethical, and organizational.

From an industry perspective, organizations face practical decisions about investment, adoption, and governance over collaborative work environments. Employers must evaluate how (and whether or not) to integrate VR tools into existing workflows, determine which collaborative activities benefit from immersion, and address questions surrounding employee well-being, inclusivity, and data governance. Moreover, industry stakeholders often lack clear frameworks for distinguishing when VR adds value to collaboration versus when it functions as a technological novelty. Without shared best practices grounded in both empirical research and lived organizational experience, adoption efforts risk fragmentation, inconsistent outcomes, and employee resistance.

This white paper addresses this need by synthesizing insights generated during the 2025 Questioning Reality conference, an interdisciplinary gathering that brought together scholars, industry leaders, and technologists to explore the future of immersive collaboration in what we refer to as collaborative work in VR. Rather than offering a definitive roadmap, this white paper articulates key tensions, emerging themes, and practical considerations identified across discussions of collaborative work in VR. By integrating the perspectives of industry leaders, scientists, and thinkers, this white paper aims to inform future research agendas and guide organizations toward more thoughtful, inclusive approaches to immersive workplace collaboration.

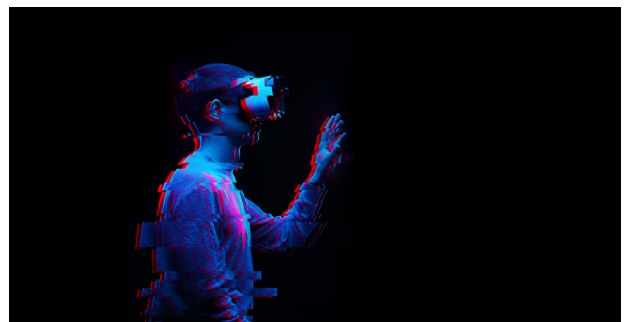
BACKGROUND ON THE QUESTIONING REALITY CONFERENCE

The Questioning Reality: Explorations of Virtual Reality conference was an in-person conference held at the Data Sciences Institute at the University of Toronto in April 2025. Co-organized by Dr. Sun Joo (Grace) Ahn, Director of the Center for Advanced Computer-Human Ecosystems (CACHE) and Professor of Advertising and Public Relations at the University of Georgia, USA, and Dr. Bree McEwan, Associate Director of the Data Sciences Institute and Associate Professor at the Institute for Communication, Culture, and Information Technology at the University of Toronto Mississauga, Canada. The conference was made possible through the generous support of the Alfred P. Sloan Foundation, a not-for-profit, mission-driven grantmaking institution dedicated to improving the welfare of all through the advancement of scientific knowledge.

Started in 2024, the invite-only QR conference brought together industry professionals, leading scholars, and enthusiasts working in the fields of virtual reality, augmented reality, extended reality, and mixed realities. Blair MacIntyre, Global Head of Immersive Technology Research at JP Morgan Chase, and Pablo Pérez, Lead Scientist at the Extended Reality Lab at Nokia, delivered stimulating keynotes. Rather than a typical academic conference where speakers present work that is already completed, QR aimed to have attendees come together at the beginning of the research process—to focus on spirited discussions, brainstorming and Blue Sky discussions, networking, and collaborative worldmaking. Core topics of the conference included generative artificial intelligence and VR as well as collaborative work and VR. The results of discussions around the latter are the focal point of this whitepaper.

COLLABORATIVE WORK IN VR: INSIGHTS, RECOMMENDATIONS, PROVOCATIONS

Uniquely situated in the conference landscape as a gathering of scholars and industry professionals around a shared excitement for VR technologies in the 21st century, QR attendees spent almost three days networking, brainstorming, and discussing issues around collaborative work and VR. The following sections offer a synthesis of key discussion strands across the conference activities, organized into work as a context for VR, avatar design, and recommendations for future directions on addressing collaborative work in VR.



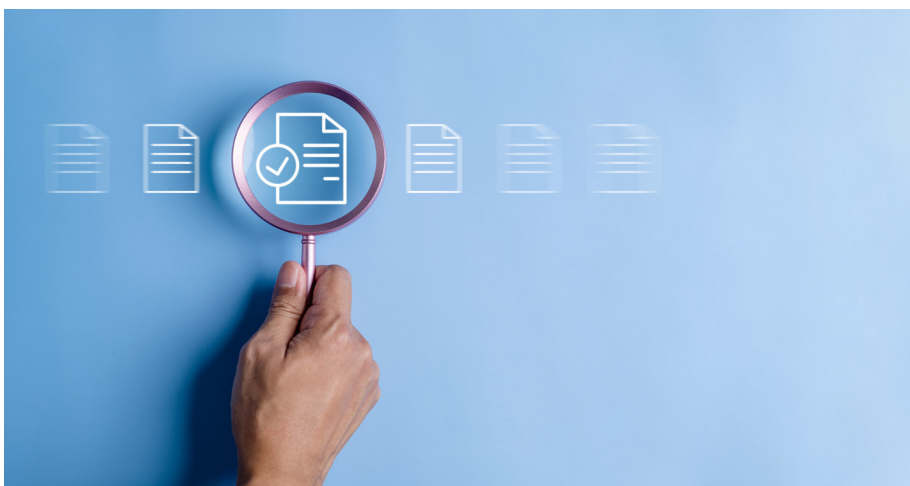
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Work as a Unique Context for VR

VR in collaborative workplaces has several implications for employees and employers alike given the unique dynamics and culture within the workplace. In particular, introducing VR into work-

places will likely face various types of adoption barriers. Most prominently are issues around trust and data privacy. VR in the workplace should protect employees' personal data, including voice and visual likeness information, as much as possible. Given that VR offers affordances that go beyond email, instant messaging, and video conferencing by affording embodiment and a sense of co-presence in virtual environments, employees may be more hesitant to participate in VR in the workplace due to concerns over data privacy. Similar to data privacy concerns around video conferencing, cloud services, and email providers, employees may be concerned about the confidentiality of VR providers, which may affect what kinds of conversations employees can have in VR environments. This likely also affects management perspectives, as organizational leaders will have to ensure confidentiality and protection of their intellectual property whenever possible. For instance, a manufacturing company may be excited about prototype design in VR yet be concerned about cybersecurity and intellectual property risks if relying on third-party VR environments and/or headsets. This raises the question: to what extent are legal, personnel, or other confidential conversations and interactions acceptable for a given organization within a third-party VR environment?

Introducing VR into the workplace will demark a change in workplace norms and customs. Given that work is a unique context of everyday life that comes with specific expectations, norms, and standards—such as professionalism, a particular way of self-presentation, interaction norms based on organizational hierarchy and role, and more—introducing VR into the workplace should be intentional and specific. Rather than moving all aspects of the workplace into VR, it might be prudent for employers to create individual spaces for specific purposes that allow segmentation and privacy. For example, supervisors may create specific VR environments for one-on-one interactions with their team members, or employees may utilize social VR rooms for mentorship purposes. In other words, introducing VR into the context should not replicate the existing context but should rather introduce added value by changing the nature of the task or its outcomes. VR might offer different benefits based on context, whereas employee-employee conversations will likely operate differently compared to employee-customer interactions in VR.



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Another adoption barrier relates directly to the employee perspective. Mental and physical health outcomes for employees are somewhat unclear, but may mirror outcomes associated with Zoom fatigue and similar reactions to prolonged engagement through technological-mediated communication channels in the workplace. Employees may need extra time to get accustomed to the technical setup of their VR headset and to

the new social scripts that will likely develop in VR workplaces. These new social scripts require a mental load to adjust, might create distrust due to the professional norms of the workplace, and

become additive to the already higher cognitive load of VR engagement. Especially given that VR is currently technically limited to its use for short-term interactions, employers should carefully weigh the costs and benefits of rolling out VR into their organizations. Accessibility and disability concerns are paramount in the workplace and include VR as well, which becomes especially important in cases where VR might be required to participate in work activities or events. Requiring employees to wear VR headsets may be policy-induced, but engaging in VR might also become socially compulsory due to social pressure and implicit messaging from coworkers and supervisors encouraging VR use. At the very least, employers should provide an option to opt out of VR use, addressing some of these employee concerns.

From a structural perspective, introducing VR into collaborative workplaces comes with several challenges. Many VR headsets may offer limited longevity due to high technical sophistication, comparatively high purchasing and maintenance costs, and planned obsolescence. Scaling a technology that is as personalized as VR may be challenging, especially given the combined need for specialized software and hardware, with the former requiring secure distribution servers and the latter relying on shipping and physical upkeep. If a hallmark of collaborative workplaces is interactive, collaborative work where colleagues and team members creatively work together, VR's mostly isolating qualities should be taken into account when considering rolling out VR in a particular organization. Given somewhat high usability barriers to conventional headsets, especially for new and novice users, employees may feel more isolated if forced to work in collaborative VR spaces. Organizations may need to be very proactive in facilitating training and workshops focused on usability, alongside providing opportunities for employees to connect with one another in VR. In addition to considerations specific to workplace culture and dynamics, avatar-related questions were featured strongly in the Blue Sky discussions at QR 2025.

Avatar Design

Avatar design emerged as a central and often contentious theme in discussions about collaborative work in VR. While avatars are frequently framed as technical interfaces or aesthetic choices, QR participants emphasized that avatar design is fundamentally a question of identity, power, and professional norms. In workplace settings, avatars can mediate perceptions of credibility, authority, authenticity, and belonging. As a result, design decisions surrounding avatars carry significant social and organizational consequences.

External Pressures on Self-Representation

A recurring theme across discussions was the role of workplace hierarchies and social expectations in shaping how individuals chose



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to represent themselves. Avatar self-expression is rarely neutral within professional contexts, such that those with greater organizational power or social capital often feel more freedom to experiment with unique or unconventional avatars, while those in lower-power positions might conform more closely to perceived norms of professionalism. In practice, this might mean that senior employees feel comfortable adopting playful or highly stylized representations, whereas junior employees feel pressure to maintain realism or conservative self-presentation.

These dynamics are reinforced by organizational expectations surrounding professionalism and appropriateness. For example, one discussion participant noted that management often prefers avatars that closely resemble employees' real-world appearance. Unique or highly stylized avatars can be perceived as inappropriate for serious or high-stakes conversations, despite the broader creative affordances of VR environments. Importantly, those with limited prior workplace VR experience may underestimate these workplace pressures, assuming that avatar freedom will translate directly into workplace practice.

Realism vs. Self-Perception

The relationship between realism and self-perception surfaced as an ethical and design challenge. On one hand, realistic avatars may increase trust and perceived authenticity; on the other, hyperrealistic representations raise concerns about manipulation and impression management, such as whether avatar design can unintentionally shape judgements of competence, authority, or likability, particularly when appearance influences trust in professional settings.

Related discussions touched on the “proteus effect” (Yee & Bailenson, 2007) in which individuals adopt behavioral characteristics associated with their avatars. Embodying a more powerful or idealized avatar may increase confidence or comfort, but it also raises questions about authenticity and psychological impact. Differences between avatar and physical appearance, such as height, age, or body proportions, may influence not only how others perceive a user but also how users perceive themselves.

Importantly, participants in the blue sky discussion also noted that perfect realism may not be necessary, or even desirable, in all contexts. People routinely use curated or outdated representations of themselves across digital platforms (e.g., outdated headshot photos or hyperaugmented photos for work), suggesting that some degree of augmentation is already socially accepted. This raises an unresolved question: in collaborative VR environments, what constitutes the “ground truth” of identity? Should professional avatars prioritize realism, self-perception, social function, or contextual appropriateness?



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Across discussions, avatar design emerged as a balance between freedom and constraint. Organizations and developers must navigate competing goals: enabling self-expression while maintaining professional norms, increasing realism without reinforcing bias, and supporting inclusivity without imposing rigid identity categories. As collaborative VR continues to develop, avatar systems will play a central role in shaping how trust, authority, and belonging are negotiated within immersive work environments.

RECOMMENDATIONS AND CONCLUDING THOUGHTS

The discussions at Questioning Reality 2025 highlighted that collaborative work in VR is not primarily a technical challenge but a social and organizational one as well. While immersive technologies can offer new possibilities for co-presence, creativity, and remote collaboration, their success in workplace contexts will depend on intentional design decisions that account for employee experience, organizational structures, and the social meaning embedded in avatar-based interaction. Based on the themes synthesized throughout this white paper, we offer the following recommendations to guide future development, implementation, and research.

First, organizations should avoid treating VR as a universal replacement for existing communication tools. Instead, the use of VR (among other communication technologies) should be selected according to the intended task. Immersive collaboration, in particular, should be introduced selectively for tasks that benefit from a sense of embodiment, spatial interaction, or shared environmental contexts. Employers should identify specific use cases—such as collaborative design, training simulations, mentoring, or team-building activities—where VR provides clear added value relative to video conferencing or other platforms. This purpose-driven adoption can help reduce the risk that VR is perceived as purely a novelty technology, in addition to aligning employee expectations for interaction in immersive environments according to the intended task. In addition to being intentional around the use of VR for collaboration, organizations should ensure that participation in VR environments remains flexible rather than compulsory whenever possible. Policies should explicitly address accessibility accommodations, options for opting out, and alternatives for employees who may face physical, cognitive, or social barriers to participation.

Second, because VR systems have the potential to collect a wide range of personal data, organizations must develop transparent policies regarding data use and storage. Employees should understand what is being recorded, how data are used, and who has access to those data. In parallel, organizations should proactively establish shared norms for professional conduct in immersive spaces, including expectations surrounding avatar use. Implementing such policies and norms could help reduce uncertainty around using VR technology for collaborative work and help normalize communication practices that optimize VR-mediated communication.

Third, avatar design should be treated as a core organizational and ethical consideration for collaborative work in VR, rather than a secondary aesthetic feature. Developers and organizations should prioritize customization options that support diverse identities, including age, race, body types, and disability representation. Systems should allow for a range of self-representation

styles (e.g., realistic vs. anime/ cartoonish) while also recognizing that hierarchies may influence how freely individuals feel able to express themselves. Doing so can help reduce representation gaps and support a stronger sense of belonging among employees and collaborators.

Fourth, we recommend strengthening collaborations between research and industry. The QR discussions underscored the importance of ongoing dialogue between scholars and practitioners. Industry innovation often moves faster than academic research. Collaborative partnerships through pilot studies, iterative testing, and shared evaluation frameworks can help organizations make evidence-informed decisions while advancing research agendas that reflect real-world challenges.

In conclusion, collaborative work in VR represents both an opportunity and a moment of transition in how organizations conceptualize communication, presence, and identity in professional environments. Immersive technologies have the potential to expand the boundaries of remote collaboration; however, their success (and adoption) will depend on how thoughtfully they are integrated into existing social and organizational systems. Moving forward, organizations, designers, and researchers should work together to ensure that immersive collaboration environments are effective, inclusive, human-centered, and responsive to the diverse ways people communicate and work together.

REFERENCES

Doering, L. & Tilcsik, A. (2025, October 27). Remote work reduced gender discrimination—returning to the office may change that. *The Conversation*. <https://theconversation.com/remote-work-reduced-gender-discrimination-returning-to-the-office-may-change-that-265945>

Shumway, J. (2026, January 21). As companies call employees back to the office, can some workers hold onto working from home? *CBS News*. <https://www.cbsnews.com/pittsburgh/news/working-from-home-and-companies-wanting-workers-back/>

Stechyson, N. (2025, September 5). Is remote work gone for good? More Canadians are commuting—whether they want to or not. *Canadian Broadcasting Corporation*. <https://www.cbc.ca/news/canada/remote-work-over-1.7619112>

Paradis, T. & Altchek, A. (2025, October 19). Remote companies like Dropbox and Atlassian say they're getting flooded with applications, as other firms RTO. *Business Insider*. <https://www.businessinsider.com/remote-companies-say-getting-flooded-with-applications-2025-10>

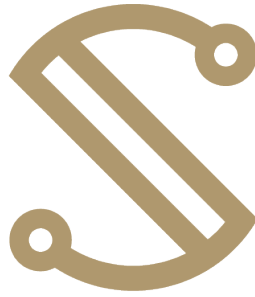
Parker, K. (2025, January 13). Many remote workers say they'd be likely to leave their job if they could no longer work from home. *Pew Research*. <https://www.pewresearch.org/short-reads/2025/01/13/many-remote-workers-say-theyd-be-likely-to-leave-their-job-if-they-could-no-longer-work-from-home/>

Yee, N., & Bailenson, J. (2007). The Proteus effect: The effect of transformed self-representation on behavior. *Human communication research*, 33(3), 271-290. <https://doi.org/10.1111/j.1468-2958.2007.00299.x>

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CONTRIBUTORS

Farzan Rahimi, MacEwan University, Canada
Robby Ratan, Michigan State University, USA
Raheleh Saryazdi, Trent University, Canada
Shima Dadkhahfard, MacEwan University, Canada
Mia Wang, University of Colorado, USA
Michelle Lui, University of Toronto, Canada
Michael Nixon, University of Toronto, Canada
Gary Bader, University of Toronto, Canada
Sho Conte, University of Toronto, Canada
Adam Frost, University of Toronto, Canada
Eugene Kukshinov, University of Waterloo, Canada
Gigi Wong, Western University, Canada
Yoon Lee, Syracuse University, USA
Miguel Barreda-Angeles, European Commission, Italy
Mark Roman Miller, Illinois Institute of Technology, USA
Sue Lim, Michigan State University, USA
Tammy Lin, National ChengChi University, Taiwan
Diego Gomez-Zara, University of Notre Dame, USA
Mary Beth Oliver, Pennsylvania State University, USA
Eugy Han, Stanford, USA
Lindsay Hahn, SUNY-Buffalo, USA
Jesse Fox, The Ohio State University, USA
Katherine Isbister, UC-Santa Cruz, USA
Marco Denhart, University of Arkansas, USA
Joomi Lee, University of Arkansas, USA
Kristine Nowak, University of Connecticut, USA
Josh Baldwin, University of Georgia, USA
Kyle Johnsen, University of Georgia, USA
Tony Liao, University of Houston, USA
Christopher Ball, University of Illinois, USA
Breigha Adeymo, University of Illinois-Chicago, USA
Anna Queiroz, University of Miami, USA
Danny Pimental, University of Oregon, USA
Kuo-Ting Huang, University of Pittsburgh, USA
Matthew Craig, University of Tennessee, USA
Paul Alexander, University of Toronto, Canada
Luke Stark, Western University, Canada
Blake Markel, University of Toronto, Canada
Blair MacIntyre, JP Morgan Chase, USA
Pablo Perez, Nokia, Spain
Matthew Wren, VRAR Chicago, USA
Maxwell McGee, ENGAGE, USA
Jie Li, EPAM Systems, The Netherlands
Jamie She, Independent, USA
Karen Alexander, XR Women, USA



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